ECHOZY: A WEB-BASED MULTILINGUAL AUGMENTATIVE AND ALTERNATIVE COMMUNICATION PLATFORM FOR STROKE PATIENTS WITH COMMUNICATION IMPAIRMENTS

UNIMAS UNIVERSITI MALAYSIA SARAWAK

FCSIT FYP SYMPOSIUM

Introduction

Echozy is a web-based multilingual Augmentative and Alternative Communication (AAC) platform designed to help stroke patients with communication impairments express their needs, emotions, and daily messages. It supports visual communication cards, bilingual content, text-to-speech, caregiver customization, and healthcare monitoring in one accessible platform.

Problem Statement

Stroke patients with speech difficulties may struggle to express basic needs and emotions. Existing AAC tools may lack localized Bahasa Malaysia support, patient-specific personalization, offline-friendly access, and progress tracking. Therefore, a simple and localized AAC platform is needed to support communication and rehabilitation.

Objectives

i. To develop a multilingual web-based AAC platform with touch-based, personalized, and offline-friendly features.

ii. To analyse stroke patients' communication needs and system requirements through existing AAC review and user feedback.

iii. To evaluate the platform's usability, accessibility, and effectiveness through user testing and feedback.

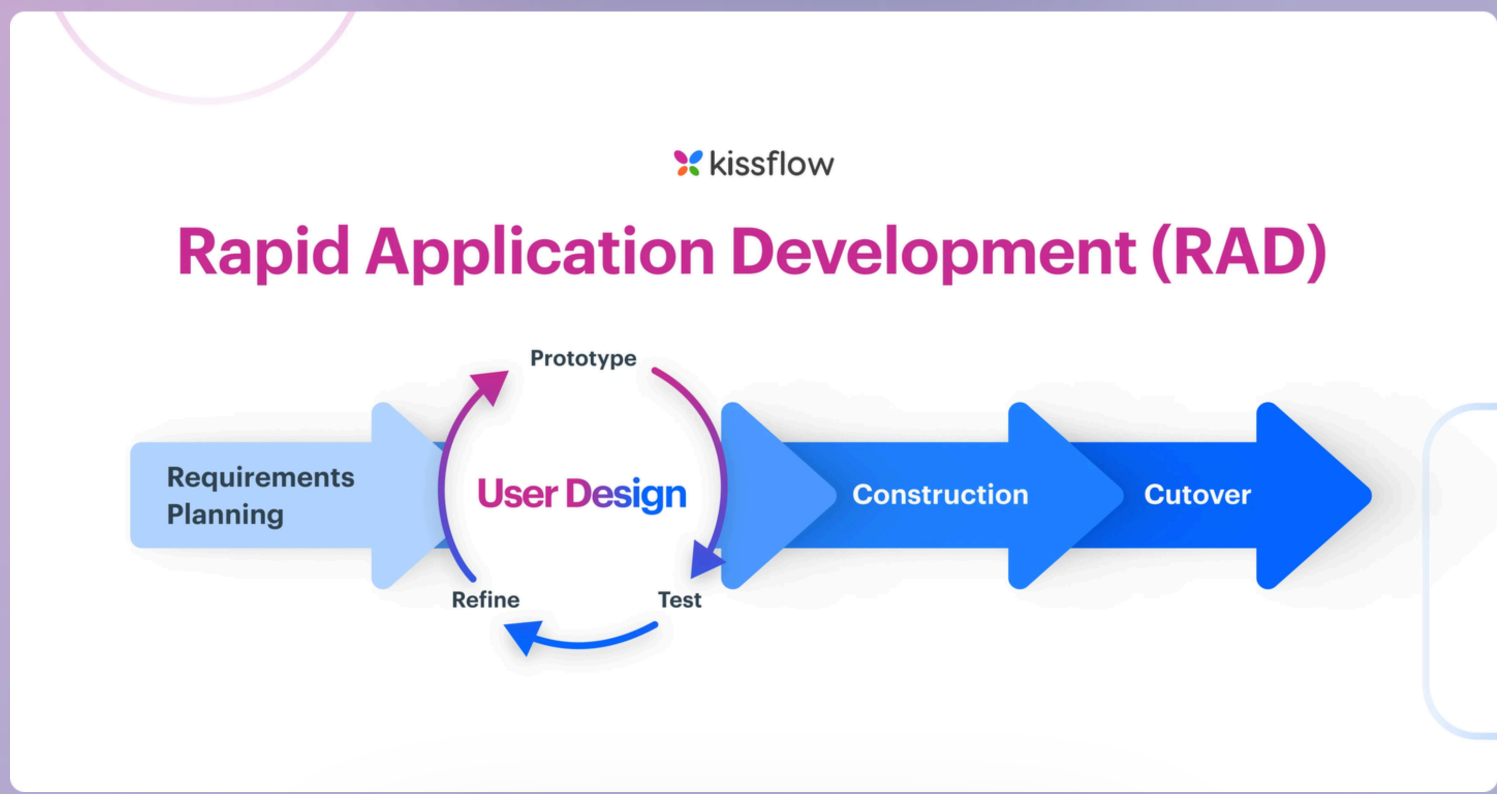
Potential Market

Echozy can be used in hospitals, rehabilitation centres, speech therapy units, elderly care centres, home-based care, and special education support environments.

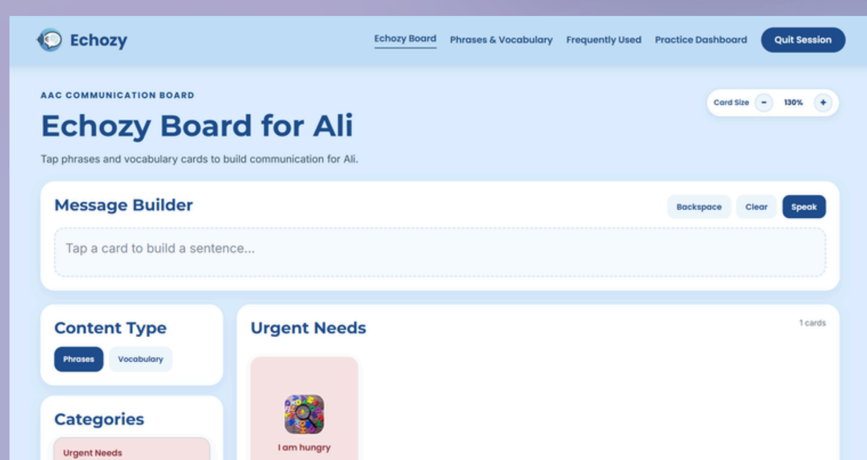
Novelty

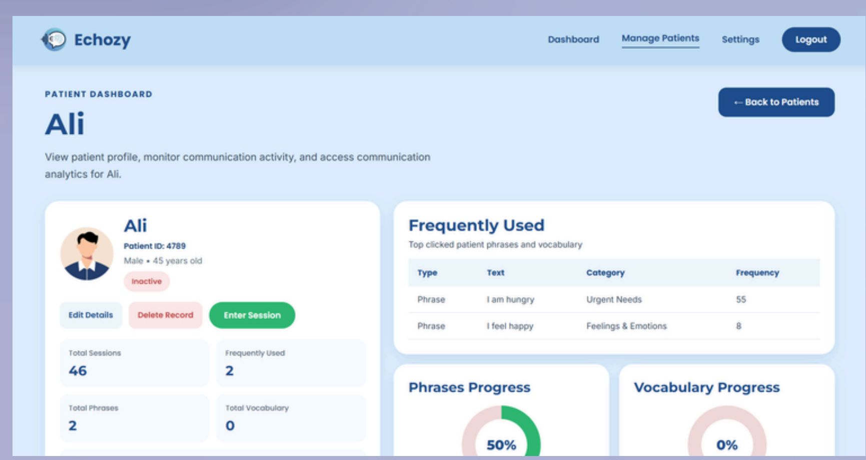
Feature / System	Echozy	CoughDrop	Proloquo2Go	Avaz AAC
Web / Cross-Platform Access	✅	✅	❌	✅
Bahasa Malaysia Support	✅	❌	❌	❌
Hybrid TTS Online + Offline	✅	❌	❌	❌
Usage-Based Personalization	✅	❌	❌	❌

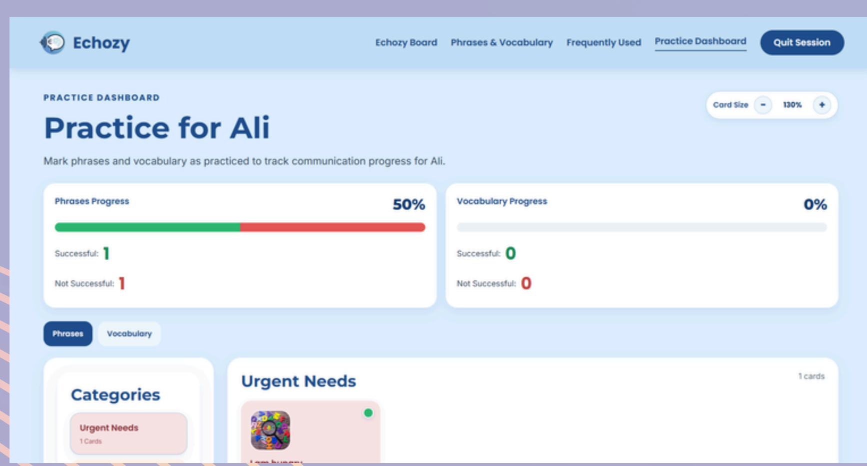
Methodology

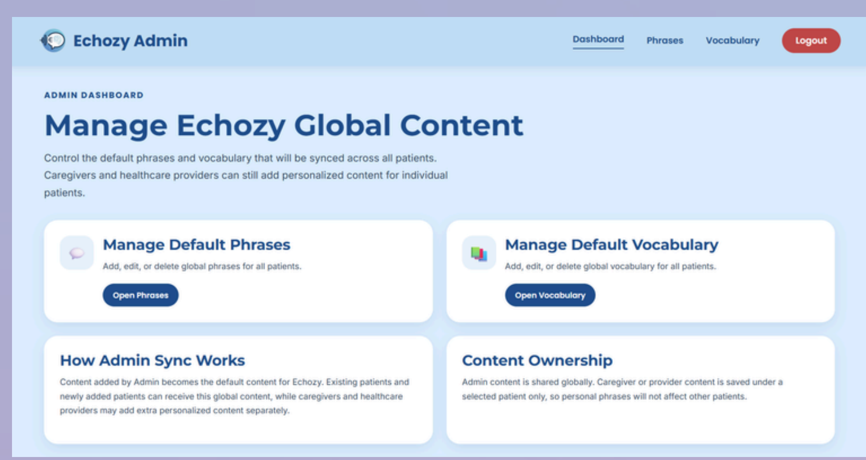


Echozy Features

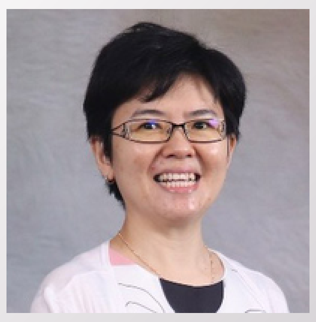








RESEARCHERS / DEVELOPERS



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